Same Deadline, Different Persistence

Submission timing and post-deadline activity among students with comparable clicks

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Page-count note. The main text ends on page 7; references, disclosures and the appendix are excluded from the 6–10 page main-text requirement.

1 Question

Online course platforms record every resource view with a timestamp. Total click volume is the obvious summary of that record and the obvious baseline for predicting who will fail, yet two students can click the same number of times and finish very differently. This project asks whether the timing of activity around a mandatory deadline carries information that the volume alone does not. The specific question is: within the same module presentation, among students whose clicks in the 14 days after the first tutor-marked assignment (TMA) deadline are comparable, are submission timing and post-deadline persistence associated with eventually failing the module rather than passing or earning a distinction? Fail is the positive class, so the measure of interest is the average precision (AP) of Fail predictions against the Fail share of the same test records. Final Withdrawn records are a separate outcome: they leave the binary comparison and return only in the at-risk description of Section 6. Three choices define the design. First, the anchor is the deadline d of each presentation's earliest TMA with positive weight, not the start of the course, because the deadline is the moment at which a large share of students either submit or disappear. Second, the observation window is symmetric: 14 days before $(d - 13 \dots d)$ give a starting-point baseline, and 14 days after $(d + 1 \dots d + 14)$ give the behaviour of interest. Third, the comparison is paired on post-deadline clicks, so a difference in persistence cannot be explained by one group simply clicking more. Everything reported here is an observational association. The pairing is constructed from the final outcome, so the resulting sample is 1:1 by design and cannot estimate how many students fail. Deadlines are the scheduled dates in the assessment table; OULAD records neither individual extensions nor mark-release dates, so a submission is only observed as early or late relative to the scheduled date, and no score is treated as known at the cutoff.

2 Data and design

The analysis uses the Open University Learning Analytics Dataset (OULAD), the public release that accompanies the module-level records of 32,593 student enrolments in seven modules and 22 presentations. The unit of analysis is one enrolment (module × presentation × student), because the

Sample-flow step	Records
all_enrolments	32593
has_first_weighted_tma	30059
registered_by_deadline	29990
registered_at_cutoff	24594
banked_excluded	469
withdrawn_at_cutoff	4319
binary_records	19806
fail_records	6092

Table 1: Ordered eligibility flow. A record must carry a first weighted TMA, be registered by the deadline, still be enrolled at $d+14$ and not be a banked attempt. Withdrawn records drop out of the binary comparison but remain available to Section 6.

Module	Presentation	d	Enrolments	Eligible	Fail	Pass/Dist.
AAA	2013J	19	383	367	43	278
AAA	2014J	19	365	329	42	245
BBB	2013B	19	1767	1511	455	798
BBB	2013J	19	2237	1774	495	1056
BBB	2014B	12	1613	1242	376	716
BBB	2014J	54	2292	1733	391	1152
CCC	2014B	32	1936	1392	375	663
CCC	2014J	32	2498	1879	400	1000
DDD	2013B	25	1303	1106	352	506
DDD	2013J	25	1938	1572	403	815
DDD	2014B	25	1228	976	246	470
DDD	2014J	20	1803	1391	333	769
EEE	2013J	33	1052	908	200	609
EEE	2014B	33	694	595	164	356
EEE	2014J	33	1188	982	189	679
FFF	2013B	19	1614	1438	419	782
FFF	2013J	19	2283	1915	486	1081
FFF	2014B	24	1500	1243	366	638
FFF	2014J	24	2365	1772	357	1101

Table 2: The nineteen analysed presentations. Deadlines fall between day 12 and day 54 of the course, so a fixed course-day cutoff would compare students at very different points of the TMA cycle; anchoring on d removes that mismatch.

same person can appear in several presentations and the deadline is a property of the presentation. Of 32,593 enrolments, 24,125 satisfy every inclusion rule; 4,319 of those end in Withdrawn, and 19,806 have a binary outcome (6,092 Fail against 13,714 Pass/Distinction), so the Fail share is 30.8%. Nineteen presentations across modules AAA–FFF carry a positive-weight TMA; module GGG has none and never enters the design. Timing features come from the daily click stream. For every record the pipeline counts clicks per relative day between $d - 13$ and $d + 14$, then derives post-deadline active days, the longest run of consecutive zero-click days, the share of post-deadline clicks in the second week, the peak seven-day share, a normalised entropy of the daily distribution, and the day of first post-deadline activity. Submission fields are masked to $d + 14$: a submission recorded later is treated as not submitted, and two indicators plus the signed distance in days form the submission block.

Tolerance	Binary	Fail	Pairs	Fail coverage	Zero-click pairs	Mean click gap	Pre-click SMD
0%	19806	6092	4118	67.6%	607	0.0000	-0.306
5%	19806	6092	4768	78.3%	607	0.0020	-0.287
10%	19806	6092	4858	79.7%	607	0.0035	-0.279

Table 3: Pairing on post-deadline clicks. The 5% row is the primary specification; 0% and 10% are sensitivities. The pre-click standardised difference is reported because pairing deliberately does not balance the pre-deadline window.

Pre-deadline indicator	Fail − Pass	95% interval
Clicks before the deadline	-20.952	[-24.818, -17.380]
Active days before	-0.955	[-1.055, -0.857]
Last-day rate lift	0.187	[0.127, 0.246]
Last-three-day rate lift	0.072	[0.022, 0.123]
Deadline-day click share	0.020	[0.013, 0.027]

Table 4: Pre-deadline context for the same pairs. Pairing fixes post-deadline volume only, so these differences survive the match and are reported rather than removed.

3 Matched comparison

Pairing runs inside each presentation. Fail records are matched 1:1 with Pass/Distinction records whose post-deadline click total is first exactly equal and then within the tolerance; positive counts must satisfy a ratio bound, and zero-click records may only pair with other zero-click records. No record is reused. The greedy order is not claimed to be globally optimal, and the realised pairing is reported so the residual imbalance can be inspected. The primary match pairs 4,768 Fail records, covering 78.3% of all Fail records, with a mean relative click gap of 0.0020. Post-deadline volume is balanced almost exactly (standardised difference -0.000), but the pre-deadline window is not: its standardised difference is -0.287, and the raw difference is -20.6 clicks per record. Fail records were already somewhat less active before the deadline, so the paired difference answers a conditional question: given similar post-deadline clicks, what else differs? Fail records concentrate their clicks closer to the deadline: the last-day rate lift is 0.187 log units higher than their matched partners, and the last-three-day lift is 0.072. The gap features point the same way: the signed relative pre/post gap is 0.142 higher for Fail, which makes sense because a smaller pre-window denominator inflates the ratio. All six indicators point the same way and none of them is close to zero. Fail records are less likely to have submitted by the cutoff and more likely to submit late; among those who submitted they submit later; after the deadline they are active on fewer days, silent for longer stretches, and less active in the second week. Holm adjustment leaves every indicator significant, so the direction of the story does not depend on which of the six is treated as primary. The descriptive pattern is steep: 75.9% of records without a submission by $d + 14$ fail, against 15.9% of those who submitted a week or more early. This is descriptive, because submission and outcome both reflect the same underlying effort, but it motivates keeping submission status in every model. Requiring pre-deadline clicks within 20% keeps only 13.9% of the main pairs (664 pairs). The active-day difference stays negative (-0.398 days), so the post-deadline pattern is not simply an artefact of pre-deadline differences, but the subset is small and its interval is wider.

Primary indicator	Pairs	Fail − Pass	95% interval	p	p_{Holm}	Reject
submitted_by_cutoff	4768	-0.1235	[-0.135, -0.112]	0.0010	0.0060	yes
submitted_after_due	4768	0.0839	[0.072, 0.096]	0.0010	0.0060	yes
relative_submission_day	3859	1.3809	[1.182, 1.568]	0.0010	0.0060	yes
post_active_days	4768	-0.3683	[-0.440, -0.295]	0.0010	0.0060	yes
post_longest_inactive_run	4768	0.3888	[0.303, 0.473]	0.0010	0.0060	yes
post_week2_share	4768	-0.0368	[-0.050, -0.024]	0.0010	0.0060	yes

Table 5: The six pre-specified paired differences, with a student-component bootstrap interval and Holm adjustment across the family. A negative value means Fail records score lower. The submission-distance indicator is only defined when both sides submitted by the cutoff, which is why its pair count is smaller. Intervals are conditional on the realised pairing.

Submission band	Records	Fail	Pass/Dist.	Withdrawn	Fail share
not_submitted_by_14	2966	1507	478	981	75.9%
early_7plus	1667	240	1270	157	15.9%
early_1to6	11959	2056	8371	1532	19.7%
due_day	5373	1558	2789	1026	35.8%
late_1to7	1641	565	610	466	48.1%
late_8to14	519	166	196	157	45.9%

Table 6: Outcome mix by submission band in the eligible cohort. The band is built from the masked submission day, so a record without a submission by $d + 14$ counts as not submitted even if it submitted later.

4 Predictive information

The pairing describes who differs at comparable clicks. A separate question is whether timing adds predictive information beyond volume and submission status. The ladder is fitted on a grouped 60/20/20 split (no student spans two parts), the threshold is chosen on the validation part, and every model is refitted on train plus validation before the test part is scored. The primary model reaches Fail AP 0.6611 against a no-skill floor of 0.3078. The ladder shows where the gain comes from: M0 already contains submission status and reaches 0.6544, the post-deadline timing block moves it to 0.6574, and the submission-distance terms add the rest up to 0.6611. Most of the predictive signal is submission status rather than the fine structure of the click sequence. A gradient-boosted alternative reaches 0.6641, within noise of the logistic ladder, so there is little evidence of a strong nonlinear pattern that the linear models miss. The threshold rule actually used is recorded per model (M0: target_precision, M1: target_precision, M2: target_precision, M3: target_precision, RF_M0: target_precision, RF_M2: target_precision). The pipeline first looks for a threshold that reaches 0.80 precision on the validation part and falls back to the maximum-F1 threshold when no such point exists, so the reported precision and recall describe a realisable operating point rather than the maximum of the precision-recall curve. With the label shuffled, the same pipeline returns Fail AP 0.3088 against a shuffled prevalence of 0.3069. A learning signal that survived shuffling would indicate leakage; it does not. Moving the cutoff from $d + 1$ to $d + 14$ raises the primary model's AP from 0.6449 to 0.6604 with the same feature list. The gain is therefore information gained by observing longer, not a better model.

Pre-click tolerance	Indicator	Pairs	Share of main	Fail − Pass
20%	submitted_by_cutoff	664	13.9%	-0.081
20%	submitted_after_due	664	13.9%	0.071
20%	relative_submission_day	580	13.9%	0.979
20%	post_active_days	664	13.9%	-0.398
20%	post_longest_inactive_run	664	13.9%	0.346
20%	post_week2_share	664	13.9%	-0.063
50%	submitted_by_cutoff	1331	27.9%	-0.056
50%	submitted_after_due	1331	27.9%	0.079
50%	relative_submission_day	1187	27.9%	1.048
50%	post_active_days	1331	27.9%	-0.322
50%	post_longest_inactive_run	1331	27.9%	0.284
50%	post_week2_share	1331	27.9%	-0.040

Table 7: Sensitivity that also requires the pre-deadline clicks to be close. The subset is selected after the fact and loses most of the sample, so it cannot replace the primary comparison; it shows whether the direction survives a stricter starting-point balance.

Indicator	Pairs	Fail − Pass
submitted_by_cutoff	4161	-0.066
submitted_after_due	4161	0.096
relative_submission_day	3637	1.436
post_active_days	4161	-0.422
post_longest_inactive_run	4161	0.446
post_week2_share	4161	-0.042

Table 8: Sensitivity restricted to pairs where both records clicked at least once after the deadline, which removes the zero-click pairs.

5 Validation design

The temporal split trains on the 2013 presentations and tests on 2014, dropping from the training side any student who also appears in 2014. Because most students take one module, this is a genuine forward check in time, but it is not an external dataset: the modules and the grading regime are the same. The leave-one-module-out folds address the other direction, generalising to a module the model has not seen, and the module composition of the test part differs a lot between folds.

6 Robustness and scope

All 5 of the indicators shown keep the same sign in every one of the eight specifications, including the two whole-window matching variants that balance a different quantity. The magnitude moves: matching on total clicks leaves less post-deadline difference, which is expected because the whole-window total is partly a pre-deadline quantity. The at-risk description is the only place where Withdrawn is modelled. Reading the binary comparison as a withdrawal warning would be wrong: those records were still enrolled at the cutoff by construction, and a model that predicts Fail among survivors is not the model that should flag a student who may leave.

Model	Features	Fail AP	No-skill AP	ROC-AUC	Brier	Cal. slope	F1
M0	7	0.6544	0.3078	0.7585	0.1658	0.909	0.472
M1	14	0.6574	0.3078	0.7606	0.1651	0.912	0.477
M2	16	0.6611	0.3078	0.7647	0.1641	0.918	0.475
M3	36	0.6608	0.3078	0.7653	0.1640	0.912	0.470
RF_M0	7	0.6541	0.3078	0.7573	0.1903	0.880	0.471
RF_M2	16	0.6641	0.3078	0.7671	0.1853	0.906	0.478

Table 9: Model ladder on the grouped main test split (single holdout at the configured seed). M0 uses pre- and post-deadline volume plus submission status; M1 adds post-deadline timing; M2 adds submission distance and is the primary model; M3 adds resource composition. M4, which adds the first-TMA score, is omitted here because the frozen configuration excludes scores from the strict models; it is reported separately only as a retrospective upper bound. Thresholds are chosen on the validation part.

Post days	Model	Fail AP	Brier
1	M0	0.6380	0.1688
1	M1	0.6380	0.1688
1	M2	0.6449	0.1671
7	M0	0.6419	0.1685
7	M1	0.6414	0.1685
7	M2	0.6465	0.1672
14	M0	0.6536	0.1658
14	M1	0.6568	0.1652
14	M2	0.6604	0.1642

Table 10: Observation-window curve on the same split. A longer window supplies more information, so AP rises; the change is not a method gain and must not be read as one.

7 Limitations

- AP is average precision for Fail, not accuracy. The no-skill reference is the Fail share in the same test records.
- Pairing is 1:1 by design and cannot estimate the population failure rate. The greedy order is not globally optimal, and the bootstrap resamples students on the realised pairs only, so re-matching uncertainty is excluded.
- The binary cohort excludes records that withdrew before the cutoff. It is not a real-time early-warning population.
- The first-TMA score contributes to the final result. Models that use it are retrospective upper bounds, and OULAD records no mark-release date.
- The OULAD splits have been inspected across several rounds of exploration. The intervals and increments are exploratory even though the protocol is now frozen.
- VLE activity types are platform categories, not content quality; `studentVle` records day-level clicks only, so study time cannot be measured.

Two design facts deserve emphasis. First, the pre-deadline window is not balanced by the primary match; the sensitivity that balances it keeps only a small share of the pairs, so the honest statement is that the comparison holds post-deadline volume fixed and reports how much the pre-deadline history still differs. Second, the gain from timing features is small. Post-deadline persistence

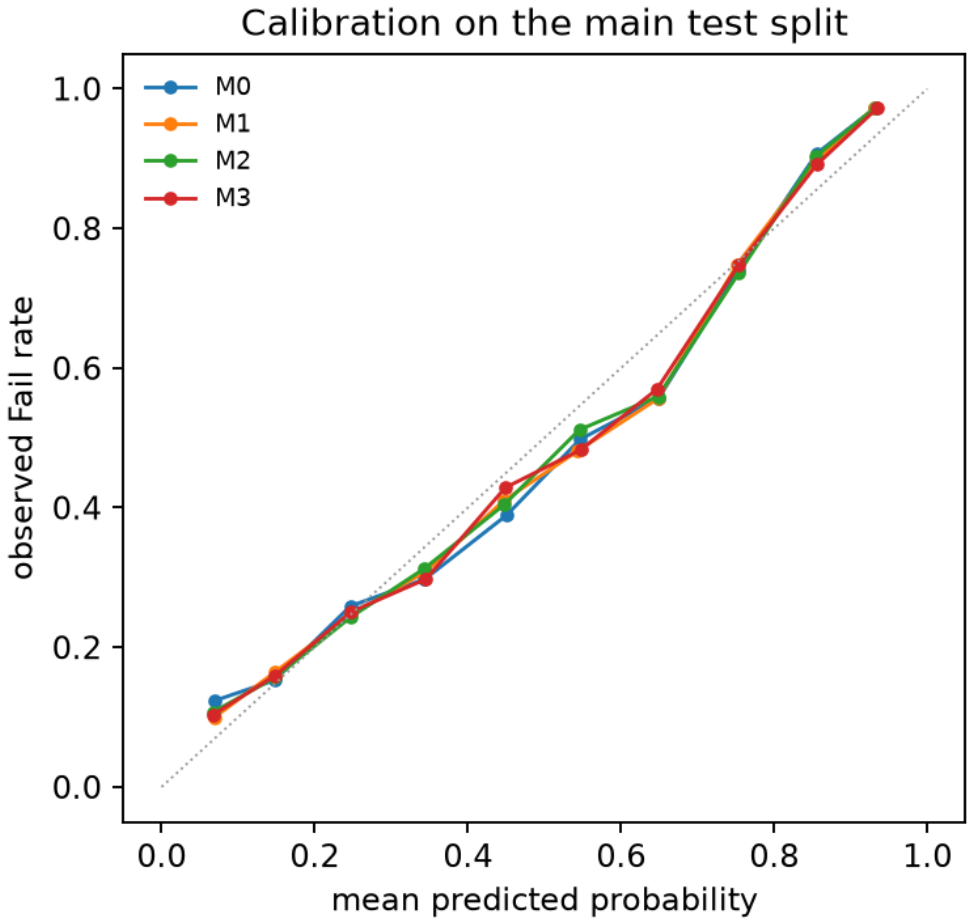


Figure 1: Reliability of the fitted probabilities on the main test split. Points near the diagonal are better calibrated; the ladder is not recalibrated after fitting, so miscalibration here is a property of the fitted model rather than a post-hoc correction.

distinguishes outcomes descriptively far more clearly than it improves prediction, and the model that improves most adds submission distance, which is itself deadline behaviour rather than a click pattern.

Disclosures

AIGC tools assisted with code scaffolding, text editing and LaTeX formatting. All analysis code, numerical results and claims were reviewed by the group, and every figure and table is regenerated by the pipeline from the public dataset. No external collaborator contributed data or results.

Data and code availability

The unmodified OULAD archive is committed with its SHA-256. Extracted CSVs, student-level features, pair indices and predictions stay outside version control. One command reproduces the

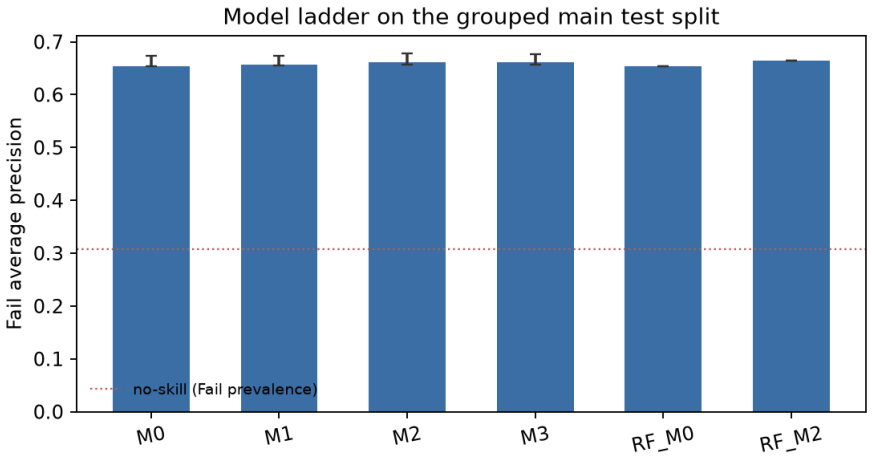


Figure 2: Model ladder on the grouped main test split; whiskers span the five grouped folds of the cross-validation arm and the dotted line is the no-skill level.

analysis; the submission ZIP contains the code, the frozen configuration and the run log.

A Exploratory extensions

Six exploratory blocks were run after the primary analysis was frozen. They do not change any primary indicator or the primary model, and each is reported here rather than in the main text. The largest residual imbalance is highest_education at -0.360 standard deviations; education, IMD band and previous attempts remain the least balanced covariates. Restricting the pairs to those that also agree exactly on education and IMD band keeps 3.1% of them, which is too small to replace the primary comparison. On that surviving subset the active-day difference is -0.120 days with a much wider interval, so the direction holds but the precision does not. Both indicators keep their sign and their magnitude under re-matching, and the re-matched intervals are only slightly wider than the fixed-pair ones. Pairing choice is therefore not doing the work here. Of the 2,966 records without a submission by $d + 14$, 33.1% eventually withdraw and 50.8% fail. Non-submission is therefore a stronger signal of leaving than of failing, which is exactly why the binary design excludes Withdrawn. Because Withdrawn is a positive case here, the AP level is much higher than in the binary design; the two numbers answer different questions and must not be placed in the same table. Inside the busiest cluster the active-day difference is -0.613 days, against -0.242 in the low-volume cluster: the persistence gap is not an artefact of heavy clickers. Adding the gap moves AP from 0.661077 to 0.661145, a change of +0.000068. The paired difference is negligible, which matches the earlier finding that the gap's descriptive content is already carried by the pre- and post-window totals.

Split	Train	Test	Train Fail	Test Fail	Student overlap
main_train_test	11892	3944	30.7%	30.8%	0
main_train_val	11892	3970	30.7%	30.9%	0
temporal	8162	11028	33.2%	29.4%	0
group_kfold_0	15844	3962	31.0%	29.7%	0
group_kfold_1	15845	3961	30.8%	30.7%	0
group_kfold_2	15845	3961	30.6%	31.3%	0
group_kfold_3	15845	3961	30.7%	31.0%	0
group_kfold_4	15845	3961	30.7%	31.0%	0
lomo_AAA	19196	608	31.3%	14.0%	0
lomo_BBB	14362	5439	30.5%	31.6%	0
lomo_CCC	16216	2438	31.5%	31.8%	0
lomo_DDD	15512	3894	30.1%	34.3%	0
lomo_EEE	17067	2197	31.1%	25.2%	0
lomo_FFF	14294	5230	30.6%	31.1%	0

Table 11: Every split is student-disjoint: no student appears on both sides. The overlap column is zero everywhere by construction and is checked at run time. The leave-one-module-out folds number six, not seven, because only six modules have a positive-weight TMA; each fold also drops the students who appear in the held-out module, so the training side shrinks.

Seed	Model	Test records	Fail share	Fail AP	Brier
42	M0	3942	0.3128	0.6659	0.1638
42	M1	3942	0.3128	0.6682	0.1632
42	M2	3942	0.3128	0.6715	0.1623
143	M0	3967	0.3166	0.6735	0.1639
143	M1	3967	0.3166	0.6755	0.1634
143	M2	3967	0.3166	0.6792	0.1625
144	M0	3958	0.3052	0.6539	0.1641
144	M1	3958	0.3052	0.6580	0.1629
144	M2	3958	0.3052	0.6572	0.1634
145	M0	3948	0.3057	0.6602	0.1627
145	M1	3948	0.3057	0.6629	0.1623
145	M2	3948	0.3057	0.6657	0.1616
146	M0	3951	0.3088	0.6778	0.1593
146	M1	3951	0.3088	0.6798	0.1586
146	M2	3951	0.3088	0.6835	0.1581

Table 12: Replication of the published two-week ladder: an 80/20 grouped holdout with the seed set of the exploratory study and no validation part. These values are the direct comparison point for the earlier tables; the main split above answers a slightly different question with a different partition.

Specification	Post days	Tolerance	Matched on	Pairs	Fail coverage	Pre SMD
primary	14	5%	post_clicks	4768	78.3%	-0.287
tolerance_0	14	0%	post_clicks	4118	67.6%	-0.306
tolerance_10	14	10%	post_clicks	4858	79.7%	-0.279
window_21_post	21	5%	post_clicks	4621	75.9%	-0.215
window_28_post	28	5%	post_clicks	4391	72.1%	-0.152
window_21_total	21	5%	total_clicks	4627	76.0%	0.051
window_28_total	28	5%	total_clicks	4477	73.5%	0.077
dedupe_events	14	5%	post_clicks	4779	78.4%	-0.268

Table 13: Robustness curve. The +21/+28 variants appear twice: matched on post-deadline clicks, which is comparable with the main design, and matched on whole-window clicks, which is comparable with the earlier +21/+28 study. The two answer different questions and must not be ranked against each other.

Indicator	Specifications negative	Specifications	Share negative
post_active_days	8	8	100.0%
post_longest_inactive_run	0	8	0.0%
post_week2_share	8	8	100.0%
submitted_after_due	0	8	0.0%
submitted_by_cutoff	8	8	100.0%

Table 14: Sign stability of each indicator across the eight specifications. A direction that flipped between reasonable choices would have to be reported as unstable; these do not flip.

Cluster	Records	Post clicks	Active days	Longest gap	Submitted by $d + 14$	Fail share
0.0	745	143.0	7.56	3.10	97.4%	47.0%
1.0	507	32.5	2.58	7.47	89.0%	55.6%
2.0	348	31.4	2.36	8.13	90.2%	48.3%

Table 15: Outcome-blind timing clusters inside the paired sample. k and the feature list were chosen without the final result. The largest cluster is also the most active, so cluster membership still tracks volume; the clusters are descriptive and are not a finding about learning styles.

Final outcome in the at-risk set	Records	Share
Fail	6092	25.3%
Pass	13714	56.8%
Withdrawn	4319	17.9%

Table 16: The at-risk set keeps Withdrawn as its own class. It is a different population from the binary comparison and answers a different question.

Covariate	Kind	SMD before	SMD after	$\lvert\text{SMD}\rvert \leq 0.10$
gender	categorical	-0.056	-0.056	yes
age_band	categorical	-0.168	-0.043	yes
highest_education	categorical	-0.331	-0.360	no
imd_band	categorical	0.180	0.161	no
disability	categorical	0.072	0.072	yes
num_of_prev_attempts	numeric	0.261	0.195	no
studied_credits	numeric	0.075	0.005	yes

Table 17: Pre-entry covariates of the matched pairs. Pairing was designed to balance post-deadline clicks only, so a remaining difference here is expected and is reported rather than corrected.

Indicator	Fixed-pair estimate	Re-matched mean	Re-matched 95% interval	Replicates
post_active_days	-0.368	-0.399	[-0.455, -0.347]	20
post_longest_inactive_run	0.389	0.412	[0.353, 0.476]	20

Table 18: Re-matching uncertainty: each replicate resamples students, rebuilds the cohort and re-runs the greedy pairing, so the interval includes the pairing itself rather than holding it fixed. Replicates are few, so it is a check on width, not a definitive interval.

Indicator	Random-effects pooled	95% interval	τ	I^2
post_active_days	-0.396	[-0.489, -0.304]	0.205	99.6%
post_longest_inactive_run	0.387	[0.312, 0.462]	0.166	99.5%
post_week2_share	-0.033	[-0.046, -0.020]	0.029	99.4%

Table 19: Per-presentation differences summarised with a DerSimonian-Laird random-effects model. A high I^2 means the presentations do not share one effect size, which is expected when presentations have different cohorts and deadlines.

Outcome	Records	Share of non-submitters	Share of eligible cohort
Fail	1507	50.8%	6.2%
Withdrawn	981	33.1%	4.1%
Pass	478	16.1%	2.0%

Table 20: Records with no submission by $d + 14$, split by final outcome. This is the subgroup the binary comparison cannot speak to, because withdrawal and non-submission overlap here.

Model	Test records	Fail-or-Withdrawn share	AP	ROC-AUC	Brier
M0	4857	43.8%	0.7483	0.7603	0.1920
M1	4857	43.8%	0.7518	0.7634	0.1908
M2	4857	43.8%	0.7549	0.7675	0.1896

Table 21: Warning target: Fail-or-Withdrawn among records enrolled at the deadline, with Withdrawn as a positive case. The cohort and the label differ from the binary comparison, so the AP is not comparable with Section 4.

Volume cluster	Records	Mean post clicks	Mean pre clicks	Fail share
zero	2171	0.0	30.6	70.8%
low	6741	17.3	78.8	37.2%
high	10894	195.0	228.9	18.8%

Table 22: C5: outcome-blind clusters on post-deadline click volume alone. Zero-click records form their own group because silence is a separate state rather than the bottom of a continuous scale.

Volume cluster	Pairs	Fail post clicks	Pass post clicks	Pre-click SMD
zero	607	0.0	0.0	-0.766
low	2140	16.2	16.2	-0.282
high	2018	155.5	155.6	-0.242

Table 23: Pairs that stay inside one volume cluster. Pairs whose two sides fall in different clusters are dropped rather than reassigned, because that is exactly a case where the 5% tolerance spans a meaningful volume difference.

Volume cluster	Indicator	Pairs	Fail − Pass
zero	submitted_by_cutoff	607	-0.516
zero	submitted_after_due	607	0.002
zero	post_active_days	607	0.000
zero	post_longest_inactive_run	607	0.000
zero	post_week2_share	607	0.000
low	submitted_by_cutoff	2140	-0.100
low	submitted_after_due	2140	0.078
low	post_active_days	2140	-0.242
low	post_longest_inactive_run	2140	0.428
low	post_week2_share	2140	-0.048
high	submitted_by_cutoff	2018	-0.031
high	submitted_after_due	2018	0.115
high	post_active_days	2018	-0.613
high	post_longest_inactive_run	2018	0.468
high	post_week2_share	2018	-0.036

Table 24: Within-cluster paired differences. Splitting by volume shows whether the primary indicators hold at every click level or are driven by the busiest records.

Arm	Features	Fail AP	Brier
M2_frozen	16	0.661077	0.1641
M2_plus_relative_gap	17	0.661145	0.1641

Table 25: C6: the signed relative pre/post gap added to the frozen primary model. This is a sensitivity arm, not a candidate replacement: the gap is a deterministic re-expression of two columns already in the model, so it can only add a nonlinear shape.

Day pattern	Fail only	Pass only	Discordant	Odds	p_{BH}
000	319	221	540	1.44	0.0001
00	187	134	321	1.40	0.0073
20	275	230	505	1.20	0.0694
200	527	480	1007	1.10	0.1709
020	662	645	1307	1.03	0.6581
201	752	787	1539	0.96	0.4088
211	401	543	944	0.74	0.0000
112	399	550	949	0.73	0.0000
122	417	645	1062	0.65	0.0000

Table 26: E7: daily click patterns coded 0 (silent), 1 (below the record's own positive median) and 2 (at or above it), allowing one skipped day. Only pairs where exactly one side shows the pattern enter the test, which removes the volume channel. 24 of 36 patterns survive BH correction. The top six rows are the most Fail-leaning and the bottom three the most Pass-leaning.

Type run	Fail only	Pass only	Discordant	Odds	p_{BH}
CF	454	612	1066	0.74	0.0000
FC	465	624	1089	0.75	0.0000
CCF	219	329	548	0.67	0.0000
CC	658	822	1480	0.80	0.0002
FCC	257	362	619	0.71	0.0002
FFC	168	249	417	0.67	0.0005

Table 27: A2: runs of the day's dominant activity type (C content, A assessment, F forum), restricted to pairs where both records clicked at least once after the deadline so the zero-click channel is removed by construction.

B Day patterns and activity compositions

The six indicators above are aggregates. A separate question is whether the *sequence* of days carries information, and whether Fail and Pass records use different kinds of resources. Three pattern analyses were run, all inside the same 5% pairs so that click volume is held fixed, and all tested with a sign test on the pairs where exactly one side shows the pattern. The strongest Fail-leaning pattern is a run of silent days (000): it appears on the Fail side alone in 319 pairs against 221 on the Pass side (odds 1.44). The strongest Pass-leaning patterns are mixed activity runs such as 122, which appear on the Pass side up to twice as often. Read as shapes rather than as causes: a silent run and an active run are two ways of describing the same active-day difference reported in Table 5. 10 of 35 type runs survive BH correction, and every one of them leans toward Pass: there is no Fail-specific activity sequence, only Fail records *lacking* a regular content run. Alternating content and forum runs are the clearest example. This is a composition finding, not a timing one, and it does not survive as a predictive increment (Table 29). The gap is present in all four quartiles, weakest in Q1_lowest (0.151 days) and strongest in Q2 (0.560 days). The weakest quartile is the one where both sides sit close to silent, so the measurement has little room there; the difference is largest in the middle of the distribution rather than at the top, which is why it cannot be read as a heavy-clicker artefact. Two cautions apply to this whole section. First, the pattern analysis codes a record's own median as the threshold, so "2" means busy *for that student*: it is a within-person contrast, not an absolute volume band. Second, the test is a sign test on discordant pairs, which is a weak design: it cannot tell whether a pattern is a cause, a consequence or a marker of the same effort that produces the final result.

Post-click quartile	Pairs	Longest silence, Fail − Pass	95% interval
Q1_lowest	1229	0.151	[0.008, 0.288]
Q2	1160	0.560	[0.335, 0.758]
Q3	1190	0.394	[0.211, 0.584]
Q4_highest	1189	0.462	[0.331, 0.641]

Table 28: A1: the longest silent stretch, split by the pair's post-deadline click level. The difference is positive in every quartile, so it is not produced by one tail of the volume distribution.

Increment arm	Features	Fail AP
M2_frozen_baseline	16	0.66108
M2_plus_deep_frozen	24	0.66133
M2_plus_deep_deduplicated	20	0.66122
Δ deep_deduplicated_minus_baseline	–	+0.00016
Δ deep_minus_baseline	–	+0.00027

Table 29: Pre-registered increment test (protocol frozen before the run). The decision rule was an interval that excludes zero *and* a point estimate of at least 0.005; neither arm meets it, so the deep indicators add no predictive information.

C Verification of the exploratory extensions

The blocks in Appendix A are exploratory. Four checks were added afterwards to test whether any of them survives: a pre-registered increment test, a transfer summary, a subgroup audit with a minimum detectable effect, and two negative controls (delay typology and sequence ordering). None of them changes a primary number. The frozen arm moves Fail AP from 0.66108 to 0.66133, a difference of +0.00027 with a 95% interval of [-0.00039, +0.00095]. A de-duplicated arm, which removes the two columns that duplicate primary features, gives the same verdict. 1 of 10 contrasts survive BH correction: gender (all_features, gap +0.051, BH p 0.025). 7 contrasts have a gap smaller than the minimum detectable effect, so those are undetected rather than absent. Two negative controls complete the picture. The delay profiles in Appendix A cannot test the published high/low/sporadic delayer typology: the clustering runs on (submission fraction, window length) and the window length is strongly bimodal, so the profiles are close to a split on window length alone. The honest statement is that this operationalisation cannot separate delay from window length, not that the typology is wrong. Shuffling the fourteen day vectors changes Fail AP by -0.0022 (-0.0073 to 0.0031), so the order of the days carries no information beyond their multiset. This does not refute published sequence-model results, which use a different label and a different split; it says that within this design the sequence representation adds nothing.

D Reproduction

```
make test
make sim && make validate RUN_ID=$(cat .last_sim_run)
make full && make validate RUN_ID=$(cat .last_full_run)
```

Every stage writes a stage summary, a handoff note and a JSONL event; the validator checks the log against the schemas in `docs/contracts` and reports zero errors and zero warnings.

Split	Test records	Fail share	Fail AP	AP − no-skill
temporal	11028	29.4%	0.6160	0.322
lomo_AAA	608	14.0%	0.4432	0.303
lomo_BBB	5439	31.6%	0.6523	0.337
lomo_CCC	2438	31.8%	0.5997	0.282
lomo_DDD	3894	34.3%	0.6760	0.333
lomo_EEE	2197	25.2%	0.6800	0.428
lomo_FFF	5230	31.1%	0.6700	0.359

Table 30: Extrapolation summary for the primary model. Raw AP is not comparable across folds because the Fail share differs from 0.14 to 0.34, so the last column is the comparable quantity. CCC has no 2013 presentation, which makes the temporal test partly a cross-module test.

Attribute	Features	AUC gap	95% interval	MDE (80% power)	Survives BH
gender	behaviour_only	0.043	[0.009, 0.080]	0.034	no
gender	all_features	0.051	[0.017, 0.082]	0.034	yes
imd_band	behaviour_only	-0.011	[-0.051, 0.027]	0.041	no
imd_band	all_features	-0.011	[-0.047, 0.027]	0.041	no
highest_education	behaviour_only	-0.015	[-0.055, 0.019]	0.037	no
highest_education	all_features	-0.016	[-0.052, 0.022]	0.037	no
disability	behaviour_only	-0.009	[-0.063, 0.052]	0.061	no
disability	all_features	-0.009	[-0.064, 0.046]	0.061	no
age_band	behaviour_only	0.041	[-0.000, 0.078]	0.036	no
age_band	all_features	0.036	[-0.000, 0.075]	0.036	no

Table 31: Subgroup audit. A gap smaller than the minimum detectable effect means the sample cannot resolve it, which is not the same as no difference. The bootstrap p-value floor is 1/repeats, so a surviving contrast whose p sits at that floor is significant only at the resolution of the bootstrap.

Sequence arm	Features	Fail AP	ROC-AUC
mlp_ordered_daily	20	0.5472	0.7105
mlp_shuffled_daily	20	0.5495	0.7115
mlp_bag_summary	8	0.5556	0.7125
mlp_ordered_plus_M2	34	0.6550	0.7562
mlp_bag_plus_M2	22	0.6562	0.7592
M2_logistic_reference	16	0.6611	0.7647

Table 32: Ordering control on the same split. PyTorch was unavailable, so these are feed-forward networks on the ordered day vector rather than recurrent models; the contrast that matters is ordered versus shuffled.